

The Mathematics of Calibration

How every number on your calibration certificate is arrived at

A calibration certificate carries figures that look self-explanatory and are not. What does an expanded uncertainty of ± 0.98 mmHg mean? Why is the coverage factor 2.201 rather than 2? Why does a device that passed still show a test uncertainty ratio of 3.07, and why does that matter?

This handbook answers those questions from first principles. Every equation is built up from the reasoning behind it rather than quoted, worked numbers accompany each step, and section 15 carries one complete measurement from five raw readings through to the line that appears on the certificate. Section 16 is a guide to reading a certificate critically, including the checks you can do by eye.

It assumes no prior study of measurement theory. If you can add, multiply and take a square root. You can follow all of it.

1. What a calibration actually measures

When you calibrate a device, you are not measuring the device. You are measuring **the difference between what the device says and what is true.**

"What is true" is supplied by a reference standard, a simulator, a calibrator, a pressure source, whose own accuracy has been established by a higher laboratory, which in turn was established by a higher one, in an unbroken chain back to the SI definition of the unit. That chain is what the word *traceability* means on a certificate.

So a calibration produces three quantities, not one:

Quantity	Meaning
The value	The best estimate of what the device reads at this test point
The deviation	How far that estimate sits from the reference value
The uncertainty	How much the value itself could be wrong

The third is the one most often misunderstood, and it is the reason this handbook exists. An uncertainty is **not** an error. It is not a mistake anyone made. It is an honest statement of how tightly the measurement pins the answer down. A measurement without an uncertainty is not a measurement. It is an opinion with a number attached.

2. Why we take repeated readings

Apply exactly 200 mmHg to a blood-pressure module five times and you may read:

201	202	201	203	202
-----	-----	-----	-----	-----

Nothing went wrong. Real instruments scatter, because of electrical noise, thermal drift, mechanical hysteresis and the finite number of digits on the display. A single reading cannot tell you where the centre of that scatter lies, nor how wide the scatter is. Five readings tell you both.

Repeated readings therefore serve two separate purposes, and it is worth keeping them distinct in your mind:

1. **Averaging** them gives a better estimate of the true indication than any one of them.
2. **Their disagreement with each other** is itself a measurement, of how repeatable the device is.

The first gives the value. The second gives part of the uncertainty.

3. The average and the spread

3.1 The average

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

For our five readings:

$$\bar{x} = \frac{201 + 202 + 201 + 203 + 202}{5} = \frac{1009}{5} = 201.8$$

The device reads **201.8 mmHg** when 200 mmHg is applied.

3.2 The spread

We need a single number for "how much do these readings disagree?" Start with how far each reading sits from the average:

```
201 - 201.8 = -0.8
202 - 201.8 = +0.2
201 - 201.8 = -0.8
203 - 201.8 = +1.2
202 - 201.8 = +0.2
```

These sum to zero, they always do, which is what makes the average the average. So they cannot simply be added. Square them first, which removes the signs and penalises large departures more than small ones:

$$0.64 + 0.04 + 0.64 + 1.44 + 0.04 = 2.80$$

Now average the squares and take the square root to return to the original units. This is the **standard deviation**, written s:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

and for our five readings

$$s = \sqrt{\frac{2.80}{4}} = \sqrt{0.70} = 0.836660 \text{ mmHg}$$

3.3 Why divide by n - 1 and not n

This is the most-asked question in this handbook, and the answer is worth understanding rather than accepting.

You do not know the true centre of the scatter. You used the average of these same five readings as a stand-in for it. The average is, by construction, the point that sits closest to those five readings, closer than the true centre would. So the squared departures you measured are **slightly too small**, and dividing by n would report a spread slightly tighter than reality.

Dividing by $n - 1$ corrects exactly this. The intuition: with five readings you have five pieces of information, but you spent one of them working out the average, leaving four to measure the spread. Those four are the *degrees of freedom*, and they reappear in section 11.

The effect is not small at these sample sizes. With $n = 5$, dividing by 4 instead of 5 makes s about 12% larger, 0.8367 rather than 0.7483. Understating the spread would understate the uncertainty, and every subsequent number would inherit the error.

4. Type A: the uncertainty of the average

We report the average, so we need the uncertainty **of the average**, not of a single reading.

Here is the key insight. If you took another five readings you would get a slightly different average. Take five readings over and over and the averages themselves would scatter, but much less than the individual readings do because averaging cancels noise. How much less? The scatter of the average is smaller than the scatter of single readings by a factor of the square root of how many you averaged:

$$u_A = \frac{s}{\sqrt{n}}$$
$$u_A = \frac{0.836660}{\sqrt{5}} = \frac{0.836660}{2.236068} = 0.374166 \text{ mmHg}$$

This is called a **Type A** uncertainty, meaning simply: *evaluated from repeated observations made during this calibration*. That is all "Type A" means. It is a statement about where the number came from, not about what kind of uncertainty it is.

The practical consequence of the square root. To halve this component you need four times the readings, not twice. Going from 3 readings to 5 buys you a 23% reduction; going from 5 to 10 buys a further 29%; going from 10 to 20 buys only 29% more again. This is why calibration procedures settle at 5 to 10 readings. Beyond that you are spending real time for diminishing returns, and the other components in section 5 and section 6 come to dominate anyway.

5. Type B: the uncertainty from the display resolution

Suppose the device displays whole mmHg only. It reads 201. What is the true indication?

Anything from 200.5 to 201.5 would display as 201. The display has told you the true value lies in a one-unit-wide window, and, crucially, **nothing more**. Every value in that window is equally consistent with what you saw.

No amount of repeated reading fixes this. Read it a thousand times and you still only know the window. This uncertainty must be evaluated from knowledge of the instrument rather than from the data, which is what makes it **Type B**:

evaluated by means other than repeated observation.

5.1 Turning a window into a standard uncertainty

To combine this with u_A in section 7, we need it expressed the same way u_A is, as a standard deviation. So: what is the standard deviation of a quantity known only to lie somewhere in a window of width d , with no value preferred?

That is a **rectangular** (or uniform) distribution, flat-topped, because every value is equally likely, and hard-edged, because values outside are impossible. Its standard deviation is:

$$u_{\text{res}} = \frac{d}{\sqrt{12}}$$

$$u_{\text{res}} = \frac{1}{\sqrt{12}} = \frac{1}{3.464102} = 0.288675 \text{ mmHg}$$

5.2 Where the square root of 12 comes from, the derivation

This is worth doing properly, because the constant looks arbitrary and is not.

The probability square. The display told us the truth lies somewhere in a window of width d and nothing more. Centre that window on zero, so it runs from $-a$ to $+a$ with $d = 2a$. Every value in it is equally likely, and no value outside it is possible.

A probability density must enclose an area of exactly 1: the truth is

somewhere. A flat density over a window of width $2a$ is therefore a rectangle of width $2a$ and height p with

$$\text{area} = 2a \cdot p = 1 \quad \Rightarrow \quad p(x) = \frac{1}{2a} \quad \text{for } -a \leq x \leq +a$$

and $p(x) = 0$ everywhere else. That rectangle, width $2a$ height $1/2a$ is the whole of what the display has told us. Everything below follows from it.

The mean is the centre. By symmetry the expected value sits at zero:

$$\mathbb{E}[X] = \int_{-a}^{+a} x \cdot \frac{1}{2a} dx = \frac{1}{2a} \left[\frac{x^2}{2} \right]_{-a}^{+a} = \frac{1}{2a} \left(\frac{a^2}{2} - \frac{a^2}{2} \right) = 0$$

which is why the *reading itself* is the best estimate: the rounding is as likely to have been upward as downward.

The variance is where the square comes in. A standard uncertainty is a standard deviation, and a standard deviation is the square root of the mean

squared departure from the centre. Because the mean is zero, that is simply $\mathbb{E}[X^2]$:

$$\text{Var}(X) = \mathbb{E}[X^2] = \int_{-a}^{+a} x^2 \cdot \frac{1}{2a} dx$$

The constant $\frac{1}{2a}$ comes outside, and $\int x^2 dx = \frac{x^3}{3}$:

$$\text{Var}(X) = \frac{1}{2a} \left[\frac{x^3}{3} \right]_{-a}^{+a} = \frac{1}{2a} \left(\frac{a^3}{3} - \frac{-a^3}{3} \right) = \frac{1}{2a} \cdot \frac{2a^3}{3} = \frac{a^2}{3}$$

Taking the square root gives the standard deviation of the rectangle:

$$\sigma = \sqrt{\frac{a^2}{3}} = \frac{a}{\sqrt{3}}$$

From half-width to full width. The resolution d is the full step between displayed values, so $a = d/2$:

$$U_{\text{res}} = \frac{a}{\sqrt{3}} = \frac{d/2}{\sqrt{3}} = \frac{d}{2\sqrt{3}} = \frac{d}{\sqrt{4 \cdot 3}} = \frac{d}{\sqrt{12}}$$

So $\sqrt{12}$ is not a magic number. It is $2\sqrt{3}$: the **2** converts full width to half-width, and the $\sqrt{3}$ belongs to the flat shape. It is what falls out of integrating x^2 across a rectangle.

A sanity check on the size. The window is d wide, and the uncertainty it contributes is $d/3.46$ a little under a third of the window. That is the right order: the truth is somewhere in the window, so the typical departure from the middle should be an appreciable fraction of it, but smaller than the window itself. A quantity of d would claim the truth is always at an edge; a quantity of $d/100$ would claim the display is far more informative than it is.

Why a rectangle and not a bell curve. A normal distribution has tails running to infinity, which would assign non-zero probability to values outside the window. Those values are not merely unlikely. They are *impossible*. If the truth were 202.3 the display would have shown 202, not 201. The physics forbids the tails, so the density must be bounded; and with no reason to prefer the centre of the window over its edges. It must be flat across it. The rectangle is not an approximation chosen for convenience. It is what the situation actually is.

5.3 The one thing to get right

d is the **full step between adjacent displayed values**, one whole count of the last digit. A display reading in whole mmHg has $d = 1$. One reading to

0.1 mmHg has $d = 0.1$.

Enter the half-step by mistake and this component comes out half its true size. If you are unsure, the test is simple: **what is the smallest change in the input that changes the displayed number?** That is d .

6. Type B: the uncertainty of the reference standard

Your reference standard is not perfect either. Its own certificate states an uncertainty, and states it, almost always, as an **expanded** uncertainty at a coverage factor of $k = 2$, because that is the convention for reporting.

You cannot combine an expanded uncertainty with the standard uncertainties from Section 4 and section 5; they are different kinds of quantity. Divide by the same k to bring it back to standard form:

$$u_{\text{ref}} = U_{\text{ref}} / k$$

If the standard's certificate says ± 0.25 mmHg ($k = 2$):

$$U_{\text{ref}} = \frac{0.25}{2} = 0.125 \text{ mmHg}$$

Check the standard's certificate, every time. If it quotes a standard uncertainty rather than an expanded one, do not divide. You would halve a component that was already correct. If it quotes $k = 3$, divide by 3. The figure to carry forward is always the *standard* uncertainty.

7. Combining the components

We now have three standard uncertainties, all in mmHg, all expressed as standard deviations:

Component	Symbol	Value	Source
Repeatability	u_A	0.374166	The five readings
Display resolution	u_{res}	0.288675	The device's last digit
Reference standard	u_{ref}	0.125000	The standard's certificate

7.1 Add the squares, not the values

Adding them directly would give 0.787841, and would be wrong, because it assumes all three effects conspire to push the result the same way at the same moment. They are independent: the noise in the readings has nothing to do with where the display happens to round, which has nothing to do with the calibration of the standard in another laboratory last year.

Independent effects partly cancel. Sometimes one pushes up while another pushes down. The correct combination adds the **squares** and takes the root, the same "root sum of squares" that gives the diagonal of a box from its three edges:

$$u_c = \sqrt{u_A^2 + u_{res}^2 + u_{ref}^2}$$

```
u_A^2    = 0.374166^2 = 0.140000
u_res^2  = 0.288675^2 = 0.083333
u_ref^2  = 0.125000^2 = 0.015625
-----
sum      = 0.238958

u_c = sqrt(0.238958) = 0.488834 mmHg
```

This is the **combined standard uncertainty**, u_c . Note it is smaller than the direct sum (0.7878) and larger than the biggest single component (0.3742) which is exactly right.

7.2 What the squares tell you about where to improve

Because components combine as squares, **the largest one dominates completely**. Express each as a share of the total:

Component	Share of u_c^2
Repeatability	59%
Display resolution	35%
Reference standard	7%

Suppose you borrowed a reference standard ten times better, taking u_{ref} from

0.125 to 0.0125. The new combined uncertainty would be 0.4727, an improvement of 3%. Nearly no gain, for considerable trouble.

Now suppose instead you improved repeatability by a factor of two, better technique, a settled environment, a warmed-up device, taking u_A to 0.1871. The combined uncertainty falls to 0.3660, a **25% improvement**.

This is the single most useful thing in this handbook. Before trying to improve an uncertainty, square the components and find the big one. Effort spent on anything else is close to wasted. And note that the resolution component cannot be improved at all without a device that displays more digits, which is often the real ceiling on what a calibration can achieve.

7.3 What this combination assumes

Two assumptions are built in, and both are reasonable here, but you should know they are there.

Independence. The components must not be linked. If they were, if, say, the same temperature error affected both the standard and the device, the cancellation would not occur and additional terms would be required.

Common units and direct effect. Every component above is already in mmHg and affects the reading one-for-one. An influence that acts *indirectly*, an ambient temperature error affecting a pressure reading through the device's temperature coefficient, must first be converted into mmHg by multiplying by that coefficient before it can join the sum. That multiplier is called a *sensitivity coefficient*. Where all components are already in the measurand's own units, as here, every sensitivity coefficient is 1 and the arithmetic simplifies to what you see above.

8. Expanded uncertainty and the coverage factor

u_c is one standard deviation. For a bell-shaped distribution that spans only about 68% of the plausible range, roughly a one-in-three chance the truth lies outside. Too weak a statement for a certificate.

Multiply by a **coverage factor** k to widen the interval:

$$U = k \cdot u_c$$

$$U = 2 \times 0.488834 = 0.977668 \text{ mmHg}$$

The result reads: **the device indicates 201.8 mmHg $\pm$ 0.98 mmHg**, and we expect the true indication to lie in that interval about 95% of the time.

k	Approximate coverage
1	68%
1.645	90%
1.96	95%
2	95.45%
2.576	99%
3	99.73%

$k = 2$ is the near-universal convention, and the reason is practical rather than mathematical: it is close enough to 95% to serve, and it makes every laboratory's certificates directly comparable. Whatever value is used **must be stated on the certificate**, because ± 0.98 means nothing without it.

What this system does. It does not assume $k = 2$. It derives the coverage factor from the effective degrees of freedom (section 11) and treats the configured value as a floor, so k is never smaller than the laboratory has specified but widens when the sample is small. For the five readings above the derived factor is **2.201**, giving

$$U = 2.201 \times 0.488834 = 1.075924 \text{ mmHg}$$

rather than the 0.977668 that $k = 2$ would give, a 10% wider and more honest interval. The factor actually applied is printed in the k column, and the notes state that it was derived. Where you see a k above 2 on a certificate, that is the sample size being accounted for, not an error.

9. Deviation from nominal, and the tolerance test

9.1 The deviation

The reference applied 200. The device indicated 201.8. The gap is:

$$201.8 - 200.0 = +1.8 \text{ mmHg}$$

The device reads **1.8 mmHg high**.

Read the sign convention on your certificate before interpreting this column. Two conventions are in use and they differ only in sign:

Convention	Formula	For this measurement	Meaning of a positive value
Error of indication	indication - reference	+1.8	Device reads high
Correction	reference - indication	-1.8	Add this to the reading to fix it

Both are correct and both are used in practice. They are exact negatives of each other, so a column labelled with one and computed as the other inverts the meaning of every row. If the certificate's heading does not make the convention explicit, establish it from a point where you know which way the device leans before trusting the column.

9.2 The tolerance test

The tolerance is the largest deviation the device is permitted. For this parameter it is ± 3 mmHg:

$$|+1.8| = 1.8 \leq 3 \rightarrow \text{PASS}$$

9.3 What that PASS does, and does not, account for

This is the most consequential point in the handbook for anyone reading a certificate.

The tolerance test above compares the deviation with the tolerance **and takes no account of the uncertainty**, even though the uncertainty was just computed and appears on the same page.

Consider a deviation of 2.9 mmHg against a tolerance of 3.0, with an uncertainty of ± 0.98 . The deviation is inside tolerance, so the test passes. But the true deviation could lie anywhere from about 1.9 to 3.9, and a good part of that range is **outside** tolerance. The honest verdict is not "pass". It is "too close to call with this measurement".

Two standard tools address this, and section 10 and section 11 cover them. When a certificate shows a simple pass/fail against tolerance, understand what you are being told:

the measured deviation was inside the limit. Not: the device is inside the limit.

10. Two checks that make a verdict trustworthy

10.1 Is the measurement good enough to judge the device?

Compare the tolerance you are judging against with the uncertainty you are judging with. The ratio is the **test uncertainty ratio**:

$$\text{TUR} = \text{tolerance} / U$$

$$\text{TUR} = 3.0 / 0.977668 = 3.07$$

The long-established requirement is **TUR of at least 4**. At 4:1 the measurement is sharp enough relative to the tolerance that a simple pass/fail carries a false-accept risk of well under one percent, and the uncertainty can reasonably be set aside.

Our 3.07 falls short. The measurement is not quite good enough to judge this tolerance by simple comparison, and section 7.2 already showed where to look: 59% of the uncertainty is repeatability, so more readings or steadier technique is the route to a defensible verdict.

A low TUR does not mean the calibration is invalid. It means a bare pass/fail overstates what you know, and the guard band below should be applied instead.

10.2 Guard banding: shrinking the limit to absorb the doubt

If the measurement cannot resolve the tolerance cleanly, the remedy is to accept only within a **reduced** limit. Pull the acceptance limit in by the uncertainty:

$$\text{acceptance limit} = \text{tolerance} - U$$

$$\text{acceptance limit} = 3.0 - 0.977668 = 2.022 \text{ mmHg}$$

Now:

- deviation ≤ 2.022 -> **pass**, and the uncertainty cannot change that
- deviation ≥ 3.978 (tolerance + U) -> **fail**, and the uncertainty cannot save it
- between the two -> **indeterminate**: the measurement cannot decide

Our +1.8 sits below 2.022, so it passes *with the uncertainty accounted for*, a materially stronger statement than section 9.2 made.

That third outcome is the point. A guarded decision has **three** possible results, not two, and refusing to produce a verdict when the data does not support one is a feature. A device landing in the indeterminate band is not condemned; it needs a better measurement, or a decision recorded by someone with the authority to accept the risk.

Guard banding always trades one risk for another: you reduce the chance of passing a bad device and increase the chance of failing a good one. Which matters more is a clinical and commercial judgement, not a mathematical

one, which is why the rule in force must be **stated on the certificate** rather than left implicit.

What this system does. Guard banding is applied, and the Status column reports PASS, FAIL or INDETERMINATE accordingly. The rule is printed in the certificate notes, as ISO/IEC 17025 requires of any document carrying a statement of conformity. The test uncertainty ratio is printed for every test point, and any parameter falling below 4:1 is named in a measurement-capability note rather than left for the reader to spot.

11. Small samples: when $k = 2$ is not enough

Section 8 used $k = 2$ for about 95% coverage. That value assumes you know the spread well. With five readings you do not, s was itself estimated from four degrees of freedom (section 3.3), and it could easily be too small.

For small samples the correct multiplier comes from Student's t distribution which widens as the sample shrinks:

Readings n	Degrees of freedom	t for 95%
3	2	4.30
5	4	2.78
10	9	2.26
20	19	2.09
Large	,	1.96

At $n = 5$ the honest multiplier for 95% coverage is **2.78**, not 2. Using 2 understates the expanded uncertainty by roughly 40%.

The full treatment is not quite that severe, because only the Type A component carries few degrees of freedom, the resolution and reference components are effectively known, and blending all three (by a standard calculation called the Welch-Satterthwaite formula) yields an *effective* degrees of freedom higher than 4, and so a multiplier between 2 and 2.78. The correction is real but smaller than the table alone suggests.

What this system does. It computes the effective degrees of freedom by the Welch-Satterthwaite formula and looks up t at 95% for that value, then uses whichever is larger: that factor, or the one configured on the parameter. For the worked example the effective degrees of freedom is about 11.7, higher than the 4 that Type A alone would give, because resolution and the reference standard make up 41% of the budget, and the factor is 2.201.

The practical guidance:

- With **10 or more** readings the derived factor sits close to 2.
- With **3 to 5** readings it will visibly exceed 2. That is correct, not a fault.
- With **2 readings** there is one degree of freedom and t is **12.706**. A certificate showing a factor that large is telling you the measurement rests on two observations and cannot support a 95% statement. Treat it as a prompt to repeat the measurement with more readings, not as a result.
- **Never** compare an uncertainty computed at $k = 2$ with one computed from t as though they were the same quantity.

12. Linearity: is the device consistent across its range

A device can be accurate at 200 mmHg and poor at 50. Calibrating at several test points across the range reveals the shape of the relationship, not just one point on it.

Plot reference value (horizontal) against indicated value (vertical). A perfect device gives a straight line at 45°. Real devices give something near a straight line, and the informative question is **how near**.

Fit the best straight line through the points by least squares, the line minimising the total squared vertical distance to the points:

$$\text{slope} = \frac{n \mid \text{sum}(xy) - \text{sum}(x) \mid \text{sum}(y)}{n \mid \text{sum}(x^2) - (\text{sum}(x))^2}$$
$$\text{intercept} = (\text{sum}(y) - \text{slope} \mid \text{sum}(x)) / n$$

Then for each test point, the **residual** is how far the actual reading sits from that fitted line:

$$\text{residual} = \text{indicated} - (\text{slope} \mid \text{reference} + \text{intercept})$$

The largest residual, usually expressed as a percentage of full scale, is the

linearity error.

Slope, intercept and residuals each say something different, and separating them is the value of the exercise:

Feature	What it indicates	Typical remedy
Intercept away from zero	A constant offset across the whole range	Zero adjustment
Slope away from 1	A proportional (gain) error	Span adjustment
Large residuals	Genuine non-linearity	Not adjustable; a limit on the device

What this system does. The certificate prints slope, intercept, the largest deviation from the fitted line and that deviation as a percentage of span, for every parameter with two or more test points. The chart shows each parameter against its fitted line with a residual panel beneath, because at certificate scale a small non-linearity on a 45-degree line is invisible without one.

This is why linearity matters practically. Offset and gain errors are usually correctable at the device. A large residual is not. It is a characteristic of the instrument, and it tells you the device cannot be trusted to interpolate between the points you tested.

13. Drift: how the device behaves over years

One calibration is a snapshot. Several, over time, show a trajectory, and the trajectory is what determines how often the device actually needs calibrating.

Fit a straight line to deviation against elapsed time, using the same least squares arithmetic as section 12 with days on the horizontal axis. The slope is the

drift rate:

$$\text{drift per year} = \text{slope (deviation units per day)} \times 365$$

13.1 How well does the line fit

A slope alone can mislead. Two or three noisy points will always support *some* line. The companion figure is R^2 , which reports how much of the movement the straight line actually explains:

$$R^2 = 1 - \frac{\text{sum of squared residuals from the line}}{\text{sum of squared departures from the mean}}$$

R^2	Reading
Above 0.9	A clear, steady trend, genuine drift
0.5 to 0.9	A trend with real scatter, plausible, watch it
Below 0.5	Mostly noise, do not plan around this slope

Always read the drift rate and R^2 together. A large drift rate with R^2 of 0.2 is not drift; it is two scattered points and a line drawn through them. With only two calibrations R^2 is 1.0 by construction, the line passes exactly through both, and means nothing at all. Drift becomes real evidence at three points and persuasive at five.

13.2 Judging a drift rate: always against the tolerance

A drift rate is a dimensioned quantity, and a bare number carries no meaning across parameters. Is 0.4 per year large? For a pressure channel with a ± 3 mmHg tolerance it is modest. For an electrical safety leakage-current limit it could be enormous. For an infusion pump in mL/min it is negligible.

The only comparison that transfers between parameters is **against that parameter's own tolerance**:

$$\text{drift per year as a fraction of tolerance} = |\text{drift per year}| / \text{tolerance}$$

Fraction of tolerance per year	Reading	Interval implication
Under 0.1	Very stable	Interval can likely be extended
0.1 to 0.25	Normal	Current interval appropriate
0.25 to 0.5	Drifting noticeably	Consider shortening
Over 0.5	Will breach within two years	Shorten now

Treat any stability grade expressed as a bare number, with no reference to the tolerance or the unit, with caution. It cannot be comparing like with like across different parameters.

What this system does. Stability is graded on the fraction of tolerance, and that fraction is printed beside the drift rate so the grade can be checked. The grades are **Very stable**, **Normal**, **Drifting** and **Urgent** on the bands above, plus **Ungraded** where no tolerance is on record, in that case no interval advice is given, rather than a grade being guessed. The overall figure is the worst parameter's, not an average, because averaging hides the one parameter approaching its limit.

13.3 From drift rate to calibration interval

Drift is what makes a calibration interval a calculable quantity rather than a convention. If a device starts within tolerance and drifts steadily, the time until it reaches the limit is:

$$\text{time to breach} = \frac{\text{remaining margin to tolerance}}{\text{drift rate per year}}$$

A device 1.8 mmHg from a 3.0 mmHg limit, drifting 0.4 mmHg/year, has $1.2 / 0.4 = 3$ years before it breaches. A calibration interval of 12 months carries comfortable margin; one of 4 years does not.

This is how a measured interval beats a conventional one in both directions, it justifies extending the interval on stable devices and shortening it on the few that need it, instead of treating every device the same.

14. Suspect readings

Occasionally one reading in a set is plainly unlike the others:

201	202	248	203	202
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That 248 is not scatter. Something happened, a disconnected lead, a transcribed digit, a bump.

The obvious test would be "how many standard deviations from the mean is it?" and it fails, because the outlier inflates both the mean and the standard deviation it is being tested against. A single bad value hides itself.

So use measures a stray value cannot distort:

- the **median**, the middle value when sorted, which ignores how extreme the extremes are;
- the **median absolute deviation (MAD)**, the median of the distances from the median.

sorted:	201	202	202	203	248
median:	202				
distances from median:	1	0	0	1	46
sorted distances:	0	0	1	1	46
MAD:	1				

The MAD is 1, completely unmoved by the 248. Now score each reading:

$$\text{modified Z} = \frac{0.6745 \mid (\text{reading} - \text{median})}{\text{MAD}}$$

for 248:	$0.6745 \times (248 - 202) / 1$	$= 31.0$
for 203:	$0.6745 \times (203 - 202) / 1$	$= 0.67$

A reading is flagged when its score exceeds **3.5**. The 248 scores 31 and is flagged decisively; everything else scores below 1.

The 0.6745 simply rescales MAD so that, for well-behaved data, the score matches the familiar "number of standard deviations".

A flag is not a licence to delete. Investigate and record. Discard a reading only when you can identify the physical cause, and note the discard on the record. Silently deleting inconvenient readings is the most direct way to make a calibration record worthless.

One caution specific to coarse displays: if more than half the readings are identical, common when the resolution is coarse relative to the noise, the MAD is exactly zero and the score cannot be computed. Outlier detection is simply

unavailable in that case, which is the safe behaviour.

15. One measurement, start to finish

Parameter: non-invasive blood pressure, systolic

Test point: 200 mmHg | **Tolerance:** ± 3 mmHg

Reference standard: ± 0.25 mmHg at $k = 2$

Device resolution: 1 mmHg | **Coverage factor:** $k = 2$

Readings: 201, 202, 201, 203, 202

```

STEP 1, Average
  x_bar = 1009 / 5                                = 201.800000 mmHg

STEP 2, Standard deviation
  squared departures = 0.64+0.04+0.64+1.44+0.04    = 2.800000
  divide by n-1 = 4                                  = 0.700000
  s = sqrt(0.700000)                                = 0.836660 mmHg

STEP 3, Type A, repeatability
  u_A = 0.836660 / sqrt(5)                          = 0.374166 mmHg

STEP 4, Type B, resolution
  u_res = 1 / sqrt(12)                               = 0.288675 mmHg

STEP 5, Type B, reference standard
  u_ref = 0.25 / 2                                   = 0.125000 mmHg

STEP 6, Combine in quadrature
  u_A^2 = 0.140000
  u_res^2 = 0.083333
  u_ref^2 = 0.015625
  total = 0.238958
  u_c = sqrt(0.238958)                              = 0.488834 mmHg

STEP 7, Expand
  U = 2 x 0.488834                                  = 0.977668 mmHg

STEP 8, Deviation from nominal
  201.800000 - 200.000000                          = +1.800000 mmHg

STEP 9, Simple tolerance test
  |+1.8| = 1.8 <= 3.0                              -> PASS

STEP 10, Measurement capability
  TUR = 3.0 / 0.977668                              = 3.07 (below 4)

STEP 11, Guarded decision
  acceptance limit = 3.0 - 0.977668                 = 2.022 mmHg
  1.8 <= 2.022                                       -> PASS (guarded)

```

Reported result

> At an applied 200 mmHg the device indicated **201.8 mmHg**, a deviation of **> +1.8 mmHg** (reading high), with an expanded uncertainty of **± 0.98 mmHg** ($k = 2$, approximately 95%), from $n = 5$ readings. The deviation is within > the ± 3 mmHg tolerance both directly and after guard banding: **PASS**. The test > uncertainty ratio is 3.07, below the customary 4:1, the dominant contribution > being repeatability at 59% of the uncertainty budget.

That final sentence is what distinguishes a metrological report from a number. It states the verdict, the rule used to reach it, the confidence, and the limitation.

16. How to read your certificate

16.1 The results table

Column	What it is	What to check
Set value	Reference applied	Points span the range you actually use
Mean	Average of the readings	,
Std dev	Spread of the readings	Small relative to tolerance; if not, repeatability is poor
Error	Deviation of the mean from the reference	Confirm the sign convention (section 9.1)
Tolerance	Permitted deviation	Matches the specification you intend to hold the device to
TUR	Tolerance divided by the expanded uncertainty	4:1 or better; below that see section 10.1
Status	PASS, FAIL or INDETERMINATE	Guard banded, per the rule in the notes

16.2 The uncertainty table

Column	What it is
Set value	The test point
n	How many readings the mean and Type A came from
Type A	From the repeated readings, repeatability
Type B	From the display resolution
Reference	From the reference standard's own certificate
Combined	Root sum of squares of the three components
Expanded	Combined $\times k$, the figure to quote
k	The coverage factor actually applied , derived per section 11

Two arithmetic checks you can do by eye.

First, the components must reconcile. Square Type A, Type B and Reference, add them, take the root: you should recover Combined. If a component is missing from the printed table the arithmetic will not close, and you should ask for the missing column before relying on the budget, a budget that does not add up cannot be checked by anyone.

Second, Expanded should equal Combined $\times k$. If the notes assert $k = 2$ while the k column shows something else, one of the two is wrong.

16.3 Five questions worth asking about any certificate

1. **How many readings?** The uncertainty depends on n, and $n = 2$ and $n = 10$ are not comparable measurements. If n is not stated, the Type A component cannot be checked.

2. **Which k , and is it appropriate for n ?** $k = 2$ with three readings is optimistic (section 11).
3. **Which decision rule?** Simple acceptance, or guard banded? They give different answers on marginal points, and a certificate issuing a statement of conformity should say which was applied.
4. **What is the TUR?** Below 4:1, a bare pass on a marginal point is a weaker statement than it appears.
5. **Is the certificate number unique and permanent?** The number is the document's identity. It should be issued once, never reissued to a different device, and never renumbered afterwards. If you can find two certificates sharing a number, or a filed certificate whose number no longer matches the record, the traceability chain in section 1 is broken no matter how good the measurement was.

A certificate that answers all five is one you can defend. A certificate that answers none may still record a perfectly good measurement, but you cannot tell from the certificate, and neither can an assessor.

Where this system stands. Questions 1 to 4 are now answered on the page: n is printed per test point, the k column carries the factor actually applied and the notes say it was derived from the effective degrees of freedom, the decision rule is stated in full, and the TUR appears per point with a named warning below 4:1. Question 5, whether the number is unique and permanent, is answered for issuance: numbers are allocated by one authority and never invented locally. It is **not** yet answered for tamper-evidence; the QR code does not yet carry a verifiable signature, so treat the printed number as a reference rather than as proof of authenticity.

17. Notation and glossary

Symbol	Name	Meaning
n	Sample size	Number of repeated readings
$\bar{x}$	Mean	Average of the readings
s	Standard deviation	Spread of the readings about their mean
u_A	Type A standard uncertainty	Uncertainty of the mean, from repeatability
u_{res}	Resolution component	From the finite display step
u_{ref}	Reference component	From the reference standard's certificate
u_c	Combined standard uncertainty	Root sum of squares of all components
U	Expanded uncertainty	$k \cdot u_c$, the figure quoted on certificates
k	Coverage factor	Multiplier setting the confidence level
d	Resolution	Smallest step between displayed values
T	Tolerance	Largest permitted deviation
R^2	Coefficient of determination	How well a fitted line explains the data
TUR	Test uncertainty ratio	T / U , is the measurement good enough

Standard uncertainty, an uncertainty expressed as one standard deviation. The form in which components must be expressed before they can be combined.

Expanded uncertainty, a standard uncertainty multiplied by k to give a stated confidence. The form in which uncertainty is reported.

Type A / Type B, how a component was *evaluated*: Type A from repeated observations in this calibration, Type B by any other means (a specification, a certificate, the resolution, judgement). The labels describe provenance only and neither is inherently more reliable.

Traceability, an unbroken, documented chain of calibrations linking this measurement to the SI definition of the unit, each link carrying its own stated uncertainty.

Degrees of freedom, how many independent pieces of information contributed to an estimate. $n - 1$ for a standard deviation from n readings.

Guard band, the amount by which an acceptance limit is pulled inside the tolerance to absorb measurement uncertainty.

18. Where this mathematics comes from

None of it is local practice. It is the international consensus, and these are the documents that define it:

- **JCGM 100:2008**, *Evaluation of measurement data, Guide to the expression of uncertainty in measurement* (the "GUM"). The authoritative source for everything in section 3 to section 8 and section 11. Freely available from the BIPM.
- **ISO/IEC 17025:2017**, *General requirements for the competence of testing and calibration laboratories*. Clause 7.8.6 requires that where a statement of conformity is issued, the decision rule used be documented and reported, the requirement behind section 10.2 and section 16.3.
- **ILAC-G8:2019**, *Guidelines on decision rules and statements of conformity*. Guard banding and conformity decisions, section 10.2.
- **JCGM 200:2012**, the *International vocabulary of metrology* (VIM). Defines error, correction, uncertainty and traceability, and settles the terminology in section 9.1.

A calibration performed and reported along these lines is defensible anywhere in the world, because these are the same rules everywhere in the world.

This handbook describes the mathematics of the calibration measurement. Worked values are exact and may be reproduced by hand as a check.